Cohort survival

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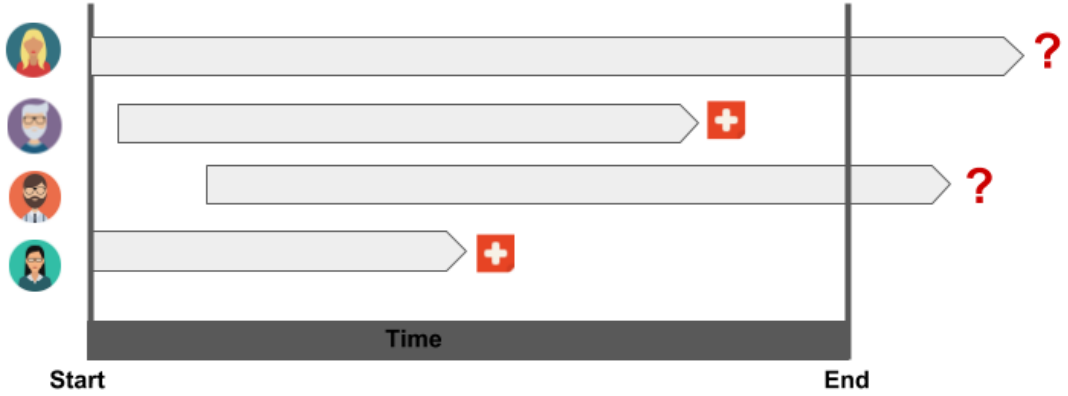
Real-World Epidemiology (RWEpi) Group

curso OMOP

Contenido

Análisis de supervivencia



Package CohortSurvival

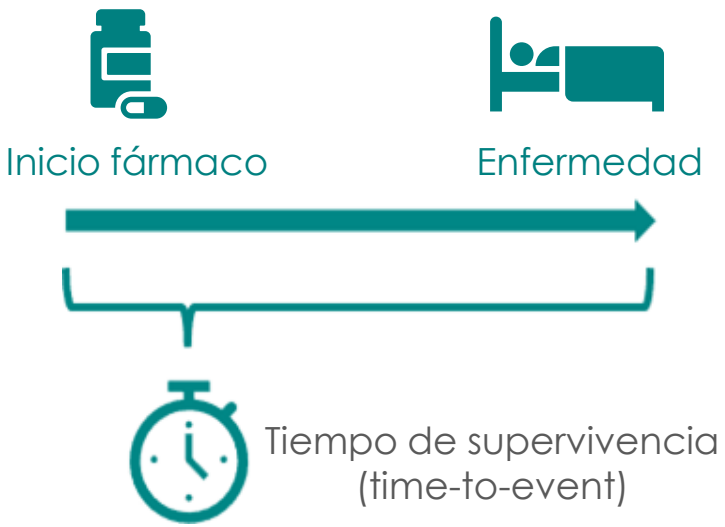
Análisis de supervivencia

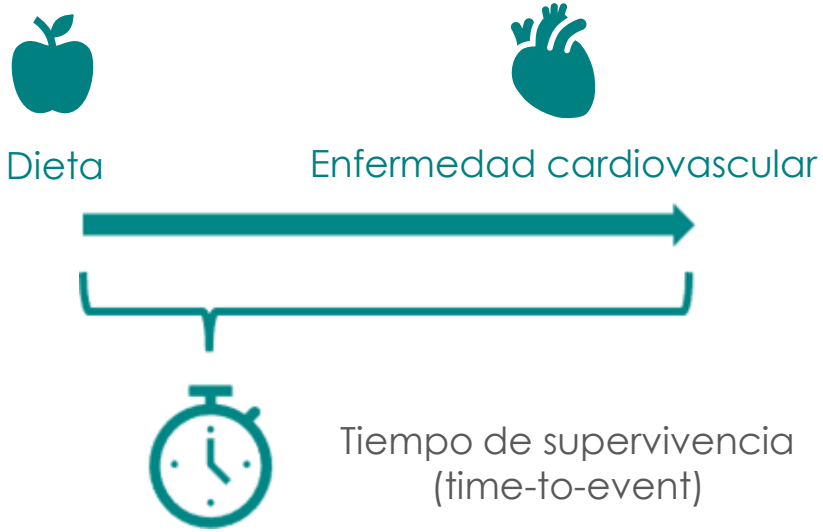
Análisis de supervivencia

- Variable de estudio es el **tiempo hasta** que se produce un **acontecimiento.**
- Tiene un tiempo de inicio y, cuando se produce un acontecimiento concreto, un tiempo final (días, semanas, meses).

Análisis de supervivencia

- **Supervivencia"** no implica que el evento de interés deba ser la **muerte**.
- El término **"análisis de supervivencia"** (o su equivalente **"análisis de tiempo hasta el fallo"**) se aplica a cualquier estudio donde se analiza el **tiempo hasta que ocurre un evento**.





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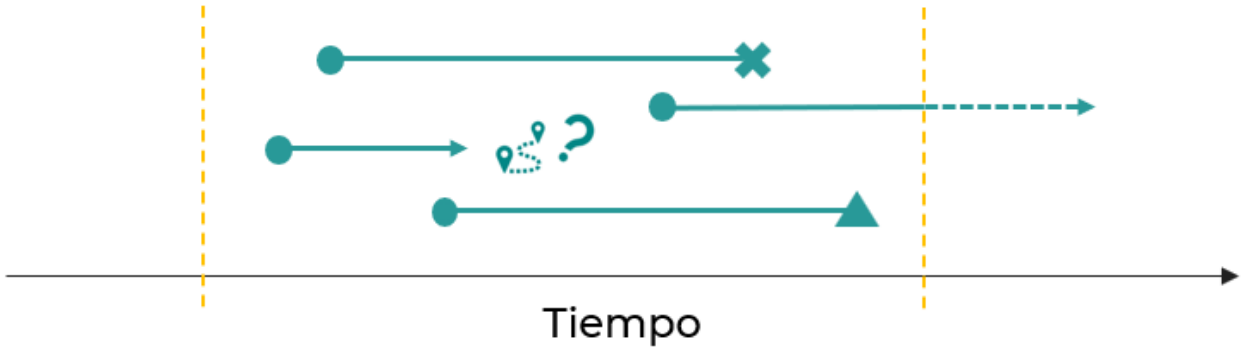
Análisis de supervivencia

En el análisis de supervivencia, estudiamos la duración desde que un participante entra en un estudio (baseline), hasta que:

- ✖ Se produce el evento de interés
- ----▸ Finaliza el estudio y el evento no ha ocurrido (aún)
- Se pierde el seguimiento del participante
- ▲ El participante experimenta un evento diferente que hace imposible un seguimiento posterior



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Análisis de supervivencia

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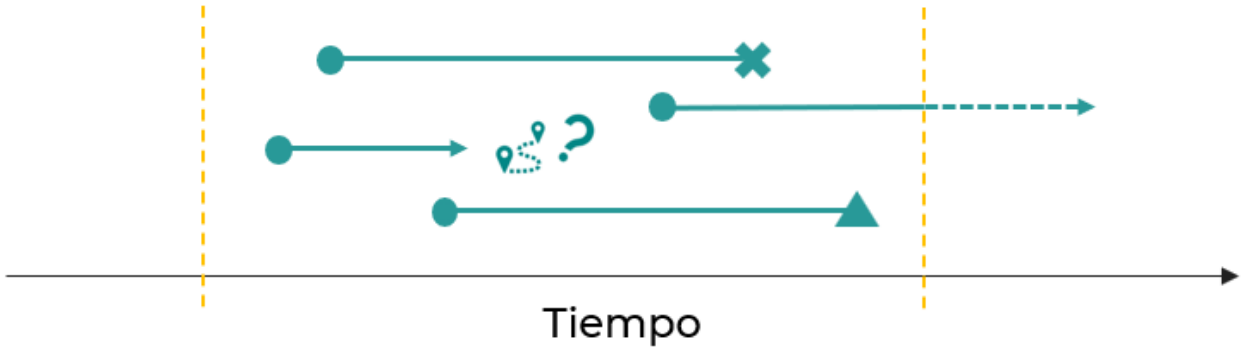
Time-to-event desconocido (censura)



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Análisis de supervivencia

En el análisis de supervivencia, estudiamos la duración desde que un participante entra en un estudio (baseline), hasta que:

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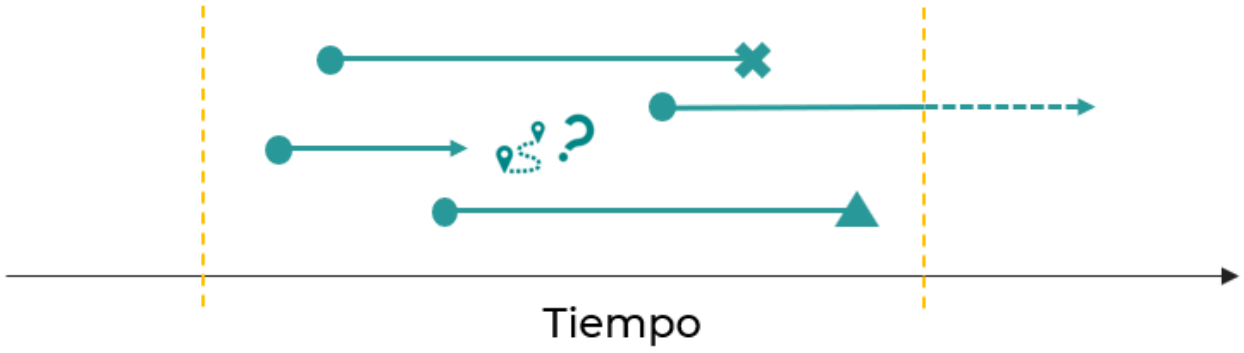
Censura administrativa



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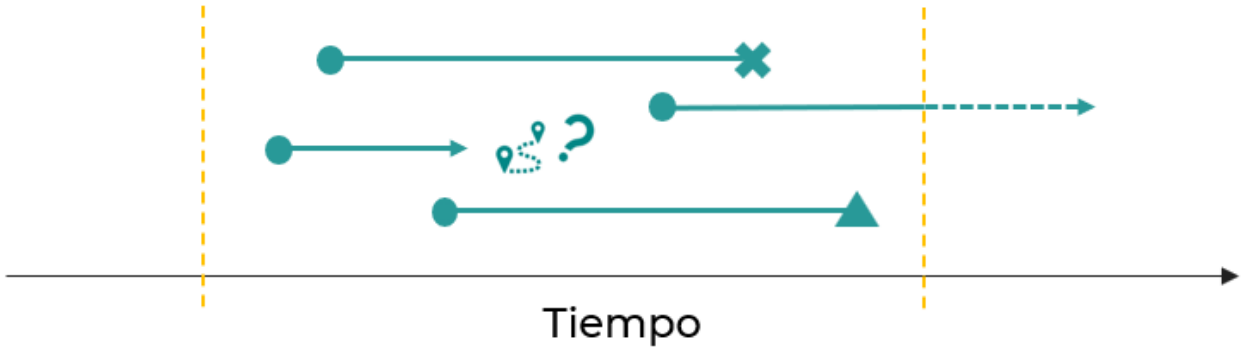
Análisis de supervivencia

En el análisis de supervivencia, estudiamos la duración desde que un participante entra en un estudio (baseline), hasta que:

- Se produce el evento de interés
- Finaliza el estudio y el evento no ha ocurrido (aún)
- Se pierde el seguimiento del participante
- **El participante experimenta un evento diferente que hace imposible un seguimiento posterior**

*Riesgo competitivo



Análisis de supervivencia

- Un **evento en competencia** es un evento (típicamente, la muerte) que impide que ocurra el evento de interés (por ejemplo, un accidente cerebrovascular).

Ejemplo: individuos que mueren por otras causas (como cáncer) ya no pueden desarrollar el evento de interés (por ejemplo, el ictus).

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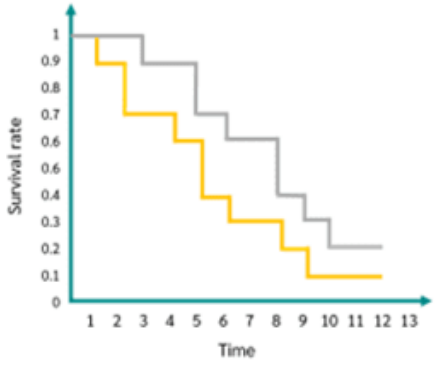
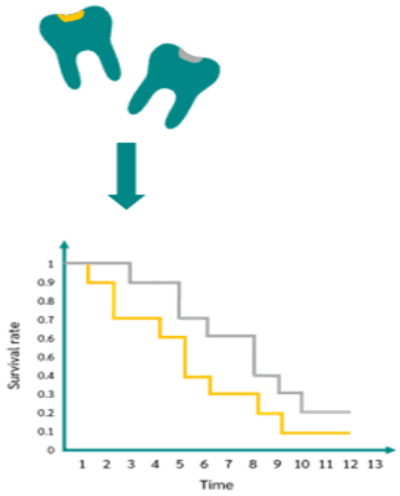
Análisis de supervivencia

Kaplan Meier Curve



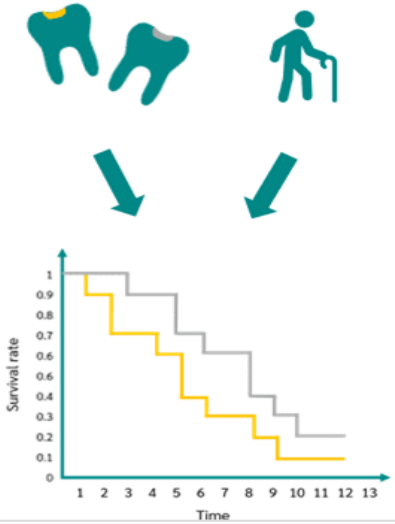
Método no paramétrico para estimar la supervivencia → muestra la probabilidad de sobrevivir (no tener el evento) a lo largo del tiempo

Log Rank Test



Prueba estadística para comparar si dos o más curvas de supervivencia son significativamente diferentes

Cox Regression



Modelo estadístico para analizar la supervivencia y evaluar cómo otras variables afectan el tiempo hasta que ocurre un evento

Package CohortSurival

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CohortSurvival 1.1.0 Reference Articles Changelog

Single outcome event of interest
Competing risk survival
Further survival analyses

CohortSurviva

CRAN 1.1.0 | codecov 100% | R-CMD-check passing | Lifecycle Stable

CohortSurvival contains functions for extracting and summarising survival data using the OMOP common data model.

Installation

You can install CohortSurvival like so:

```
install.packages("CohortSurvival")
```

Example usage

Create a reference to data in the OMOP CDM format

The CohortSurvival package is designed to work with data in the OMOP CDM format, so our first step is to create a reference to the data using the CDMConnector package.

For this example we´ll use a cdm reference containing the MGUS2 dataset from the survival package (which we transformed into a set of OMOP CDM style cohort tables). The MGUS2 dataset contains survival data of 1341 sequential patients with monoclonal gammopathy of undetermined significance (MGUS). For more information see `?survival::mgus2`.

Links
View on CRAN

License
Full license
Apache License (>= 2)

Community
Contributing guide

Citation
Citing CohortSurvival

Developers
Kim López-Güell
Author, maintainer
Edward Burn
Author
Martí Català
Author
Xintong Li
Author

https://darwin-eu.github.io/CohortSurvival/index.html

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Introducción a CohortSurvival

Developed by Edward Burn, Kim López-Güell, Marti Catala, Xintong Li, Danielle Newby, Nuria Mercade-Besora

Instalación

```
install.packages("CohortSurvival")

install.packages("devtools")
devtools::install_github("darwin-eu/CohortSurvival")
```

Librerías

```
library(CDMConnector)
library(CohortSurvival)
library(dplyr)
library(ggplot2)
library (gt)
```

Introducción a CohortSurvival

Funciones del paquete

addCohortSurvival	Add survival information to a cohort table
asSurvivalResult	A tidy implementation of the summarised_characteristics object.
availableSurvivalGrouping	Variables that can be used for faceting and colouring survival plots
estimateCompetingRiskSurvival	Estimate survival for a given event and competing risk of interest
estimateSingleEventSurvival	Estimate survival for a given event of interest
optionsTableSurvival	Additional arguments for the function tableSurvival()
plotSurvival	Plot Survival results
riskTable	Table with survival events
tableSurvival	Table with survival summary
tableSurvivalAttrition	Display the attrition of a survival result in a visual table
tableSurvivalEvents	Table with survival events

Introducción a CohortSurvival

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tableSurvivalEvents	Table with survival events

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Single Event Survival

Estimar la supervivencia para un acontecimiento de interés determinado mediante cohortes del modelo de datos comunes OMOP

```
estimateSingleEventSurvival(
  cdm,
  targetCohortTable,
  targetCohortId = NULL,
  outcomeCohortTable,
  outcomeCohortId = NULL,
  outcomeDateVariable = "cohort_start_date",
  outcomeWashout = Inf,
  censorOnCohortExit = FALSE,
  censorOnDate = NULL,
  followUpDays = Inf,
  strata = NULL,
  eventGap = 30,
  estimateGap = 1,
  restrictedMeanFollowUp = NULL,
  minimumSurvivalDays = 1
)
```

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Single Event Survival

Estimar la supervivencia para un acontecimiento de interés determinado mediante cohortes del modelo de datos comunes OMOP

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  outcomeDateVariable = "cohort_start_date",
  outcomeWashout = Inf,
  censorOnCohortExit = FALSE,
  censorOnDate = NULL,
  followUpDays = Inf,
  strata = NULL,
  eventGap = 30,
  estimateGap = 1,
  restrictedMeanFollowUp = NULL,
  minimumSurvivalDays = 1
)
```

Objeto CDM

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Single Event Survival

Estimar la supervivencia para un acontecimiento de interés determinado mediante cohortes del modelo de datos comunes OMOP

```
estimateSingleEventSurvival(
  cdm,
  targetCohortTable,
  targetCohortId = NULL,
  outcomeCohortTable,
  outcomeCohortId = NULL,
  outcomeDateVariable = "cohort_start_date",
  outcomeWashout = Inf,
  censorOnCohortExit = FALSE,
  censorOnDate = NULL,
  followUpDays = Inf,
  strata = NULL,
  eventGap = 30,
  estimateGap = 1,
  restrictedMeanFollowUp = NULL,
  minimumSurvivalDays = 1
)
```

Tabla que contiene la cohorte objetivo para el análisis de supervivencia

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Single Event Survival

Estimar la supervivencia para un acontecimiento de interés determinado mediante cohortes del modelo de datos comunes OMOP

```
estimateSingleEventSurvival(
  cdm,
  targetCohortTable,
  targetCohortId = NULL,
  outcomeCohortTable,
  outcomeCohortId = NULL,
  outcomeDateVariable = "cohort_start_date",
  outcomeWashout = Inf,
  censorOnCohortExit = FALSE,
  censorOnDate = NULL,
  followUpDays = Inf,
  strata = NULL,
  eventGap = 30,
  estimateGap = 1,
  restrictedMeanFollowUp = NULL,
  minimumSurvivalDays = 1
)
```

Tabla que contiene la cohorte de resultados o acontecimientos de interés para los cuales se quieren calcular las estimaciones de supervivencia

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Single Event Survival

Estimar la supervivencia para un acontecimiento de interés determinado mediante cohortes del modelo de datos comunes OMOP

```
estimateSingleEventSurvival(
  cdm,
  targetCohortTable,
  targetCohortId = NULL,
  outcomeCohortTable,
  outcomeCohortId = NULL,
  outcomeDateVariable = "cohort_start_date",
  outcomeWashout = Inf,
  censorOnCohortExit = FALSE,
  censorOnDate = NULL,
  followUpDays = Inf,
  strata = NULL,
  eventGap = 30,
  estimateGap = 1,
  restrictedMeanFollowUp = NULL,
  minimumSurvivalDays = 1
)
```

Identificación de las cohortes a incluir. Puede ser el cohort_definition_id o cohort_name

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Single Event Survival

Estimar la supervivencia para un acontecimiento de interés determinado mediante cohortes del modelo de datos comunes OMOP

```
estimateSingleEventSurvival(
  cdm,
  targetCohortTable,
  targetCohortId = NULL,
  outcomeCohortTable,
  outcomeCohortId = NULL,
  outcomeDateVariable = "cohort_start_date",
  outcomeWashout = Inf,
  censorOnCohortExit = FALSE,
  censorOnDate = NULL,
  followUpDays = Inf,
  strata = NULL,
  eventGap = 30,
  estimateGap = 1,
  restrictedMeanFollowUp = NULL,
  minimumSurvivalDays = 1
)
```

Estratificar

Single Event Survival

Ejemplo MGUS

El dataset MGUS2 contiene 1341 pacientes secuenciales con gammopatía monoclonal de significado incierto (MGUS)

```
cdm <- CohortSurvival::mockMGUS2cdm()
```

contiene: person, observation_period, visit_occurrence, death_cohort, mgus_diagnosis, progression, death cohort, progression type

```
── # OMOP CDM reference (duckdb) of mock ──────────────────────────────────────────────────────────────
• omop tables: observation_period, person, visit_occurrence
• cohort tables: death_cohort, mgus_diagnosis, progression, progression_type
• achilles tables: -
• other tables: -
```

Single Event Survival

Ejemplo MGUS

Esta referencia de cdm contiene tres tablas de cohortes de interés:

1) Cohorte de diagnóstico MGUS

```
cdm$mgus_diagnosis %>%
glimpse()
Rows: ??
Columns: 10
Database: DuckDB v1.4.4 [lperez@Windows 10 x64:R 4.3.2/:memory:]
$ cohort_definition_id <int> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1…
$ subject_id           <dbl> 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14,…
$ cohort_start_date    <date> 1981-01-01, 1968-01-01, 1980-01-01, 1977-01-0…
$ cohort_end_date      <date> 1981-01-01, 1968-01-01, 1980-01-01, 1977-01-0…
$ age                  <dbl> 88, 78, 94, 68, 90, 90, 89, 87, 86, 79, 86, 89…
$ sex                  <fct> F, F, M, M, F, M, F, F, F, F, M, F, M, F, M, F…
$ hgb                  <dbl> 13.1, 11.5, 10.5, 15.2, 10.7, 12.9, 10.5, 12.3…
$ creat                <dbl> 1.30, 1.20, 1.50, 1.20, 0.80, 1.00, 0.90, 1.20…
$ mspike               <dbl> 0.5, 2.0, 2.6, 1.2, 1.0, 0.5, 1.3, 1.6, 2.4, 2…
$ age_group            <chr> ">=70", ">=70", ">=70", "<70", ">=70", ">=70",…
```

Single Event Survival

Ejemplo MGUS

Esta referencia de cdm contiene tres tablas de cohortes de interés:

2) Cohorte de progresión MGUS a mieloma múltiple

```
cdm$progression %>%
glimpse()
Rows: ??
Columns: 4
Database: DuckDB v1.4.4 [lperez@Windows 10 x64:R 4.3.2/:memory:]
$ cohort_definition_id <int> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1…
$ subject_id           <dbl> 56, 81, 83, 111, 124, 127, 147, 163, 165, 1…
$ cohort_start_date    <date> 1978-01-30, 1985-01-15, 1974-08-17, 1993-0…
$ cohort_end_date      <date> 1978-01-30, 1985-01-15, 1974-08-17, 1993-0…
```

Single Event Survival

Ejemplo MGUS

Esta referencia de cdm contiene tres tablas de cohortes de interés:

3) Cohorte de muerte

```
cdm$death_cohort %>%
glimpse()
Rows: ??
Columns: 4
Database: DuckDB v1.4.4 [lperez@Windows 10 x64:R 4.3.2/:memory:]
$ cohort_definition_id <int> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1…
$ subject_id           <dbl> 1, 2, 3, 4, 5, 6, 7, 8, 10, 11, 12, 13, 14,…
$ cohort_start_date    <date> 1981-01-31, 1968-01-26, 1980-02-16, 1977-0…
$ cohort_end_date      <date> 1981-01-31, 1968-01-26, 1980-02-16, 1977-0…
```

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Single Event Survival

En este ejemplo, podemos obtener estimaciones de supervivencia para la muerte después de un diagnóstico de MGUS:

```
MGUS_death <- estimateSingleEventSurvival(cdm,
                                          targetCohortTable = "mgus_diagnosis",
                                          targetCohortId = 1,
                                          outcomeCohortTable = "death_cohort",
                                          outcomeCohortId = 1)
```

```
> MGUS_death
# A tibble: 10,767 × 13
   result_id cdm_name group_name    group_level    strata_name strata_level variable_name variable_level estimate_name estimate_type estimate_value
       <int> <chr>    <chr>         <chr>          <chr>       <chr>        <chr>         <chr>          <chr>         <chr>         <chr>
 1         1 mock     target_cohort mgus_diagnosis overall     overall      outcome       death_cohort   estimate      numeric       1
 2         1 mock     target_cohort mgus_diagnosis overall     overall      outcome       death_cohort   estimate_95C… numeric       1
 3         1 mock     target_cohort mgus_diagnosis overall     overall      outcome       death_cohort   estimate_95C… numeric       1
 4         1 mock     target_cohort mgus_diagnosis overall     overall      outcome       death_cohort   estimate      numeric       0.9697
 5         1 mock     target_cohort mgus_diagnosis overall     overall      outcome       death_cohort   estimate_95C… numeric       0.9607
 6         1 mock     target_cohort mgus_diagnosis overall     overall      outcome       death_cohort   estimate_95C… numeric       0.9787
 7         1 mock     target_cohort mgus_diagnosis overall     overall      outcome       death_cohort   estimate      numeric       0.9494
 8         1 mock     target_cohort mgus_diagnosis overall     overall      outcome       death_cohort   estimate_95C… numeric       0.9379
 9         1 mock     target_cohort mgus_diagnosis overall     overall      outcome       death_cohort   estimate_95C… numeric       0.961
10         1 mock     target_cohort mgus_diagnosis overall     overall      outcome       death_cohort   estimate      numeric       0.9386
# ℹ 10,757 more rows
# ℹ 2 more variables: additional_name <chr>, additional_level <chr>
```

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Plot Survival

Graficar los resultados de supervivencia

```
plotSurvival(
  result,
  ribbon = TRUE,
  facet = NULL,
  colour = NULL,
  cumulativeFailure = FALSE,
  riskTable = FALSE,
  riskInterval = 30,
  logLog = FALSE,
  timeScale = “days”,
  style = NULL
)
```

Resultados de supervivencia

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PlotSurvival

Graficar los resultados de supervivencia

```
plotSurvival(
  result,
  ribbon = TRUE,
  facet = NULL,
  colour = NULL,
  cumulativeFailure = FALSE,
  riskTable = FALSE,
  riskInterval = 30,
  logLog = FALSE,
  timeScale = “days”,
  style = NULL
)
```

Atributos que facilitan la visualización de los datos

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PlotSurvival

Graficar los resultados de supervivencia

```
plotSurvival(
  result,
  ribbon = TRUE,
  facet = NULL,
  colour = NULL,
  cumulativeFailure = FALSE,
  riskTable = FALSE,
  riskInterval = 30,
  logLog = FALSE,
  timeScale = “days”,
  style = NULL
)
```

TRUE or FALSE

Permite graficar como probabilidad de fallo acumulado en vez de probabilidad de supervivencia

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PlotSurvival

Graficar los resultados de supervivencia

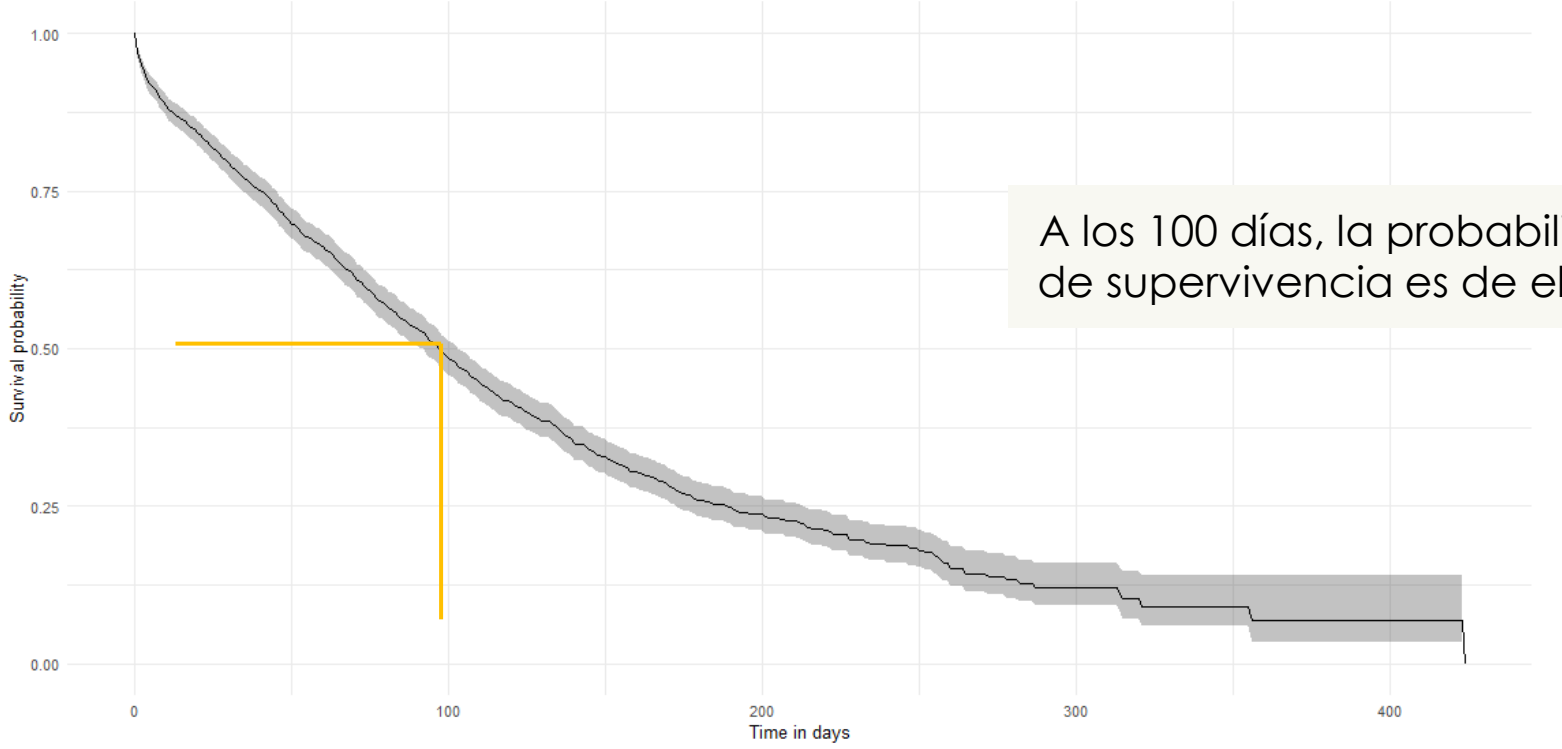
```
plotSurvival(
  result,
  ribbon = TRUE,
  facet = NULL,
  colour = NULL,
  cumulativeFailure = FALSE,
  riskTable = FALSE,
  riskInterval = 30,
  logLog = FALSE,
  timeScale = “days”,
  style = NULL
)
```

Añade una tabla de riesgo debajo del grafico de supervivencia, para el cual podemos definir el intervalo de tiempo que queramos

PlotSurvival

Ejemplo MGUS - PlotSurvival

```
plotSurvival(MGUS_death)
```

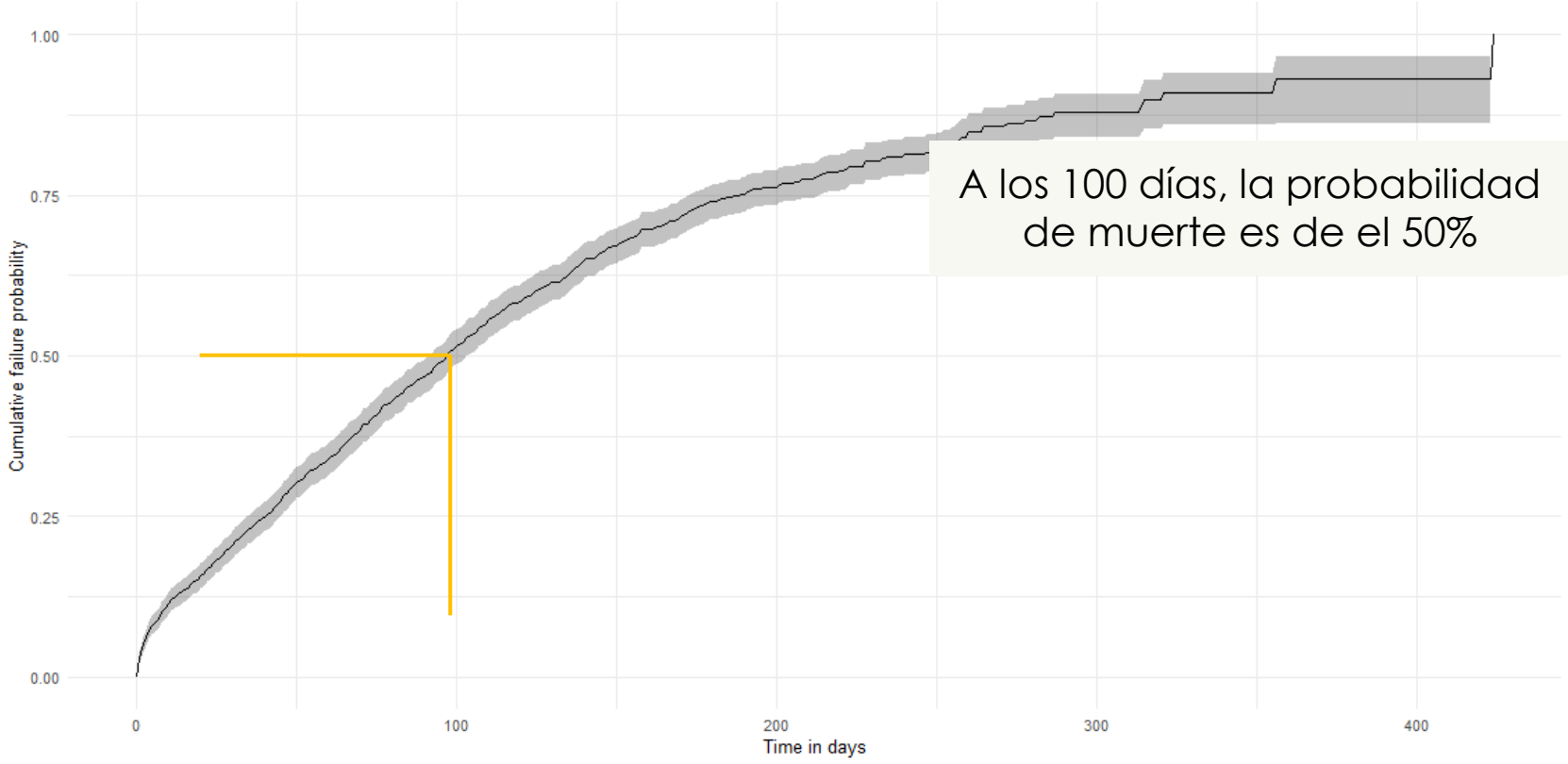


A los 100 días, la probabilidad de supervivencia es de el 50%

PlotSurvival

Ejemplo MGUS - PlotSurvival

```
plotSurvival(MGUS_death, cumulativeFailure = TRUE)
```

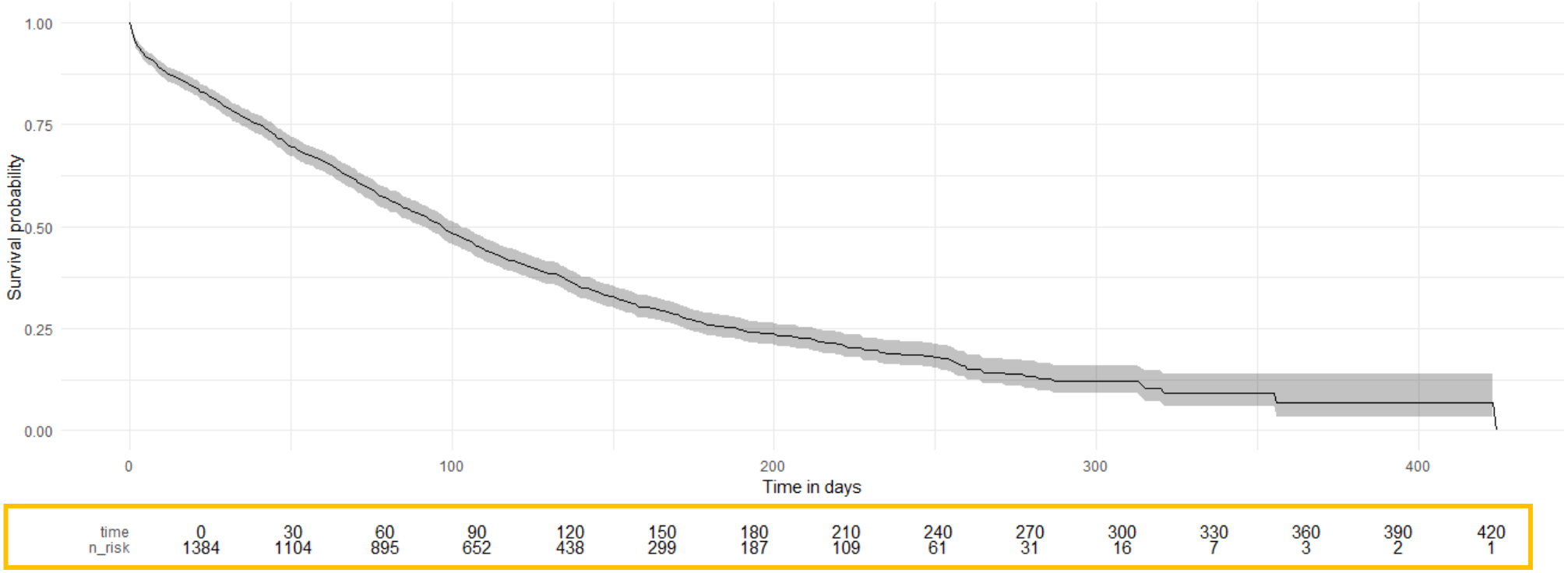


A los 100 días, la probabilidad de muerte es de el 50%

PlotSurvival

Ejemplo MGUS - PlotSurvival

```
plotSurvival(MGUS_death, riskTable = TRUE, riskInterval = 30)
```



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Stratified results

```
MGUS_death <- estimateSingleEventSurvival(cdm,
  targetCohortTable = "mgus_diagnosis",
  targetCohortId = 1,
  outcomeCohortTable = "death_cohort",
  outcomeCohortId = 1,
  strata = list(c("age_group"),
                c("sex"),
                c("age_group", "sex")))
```

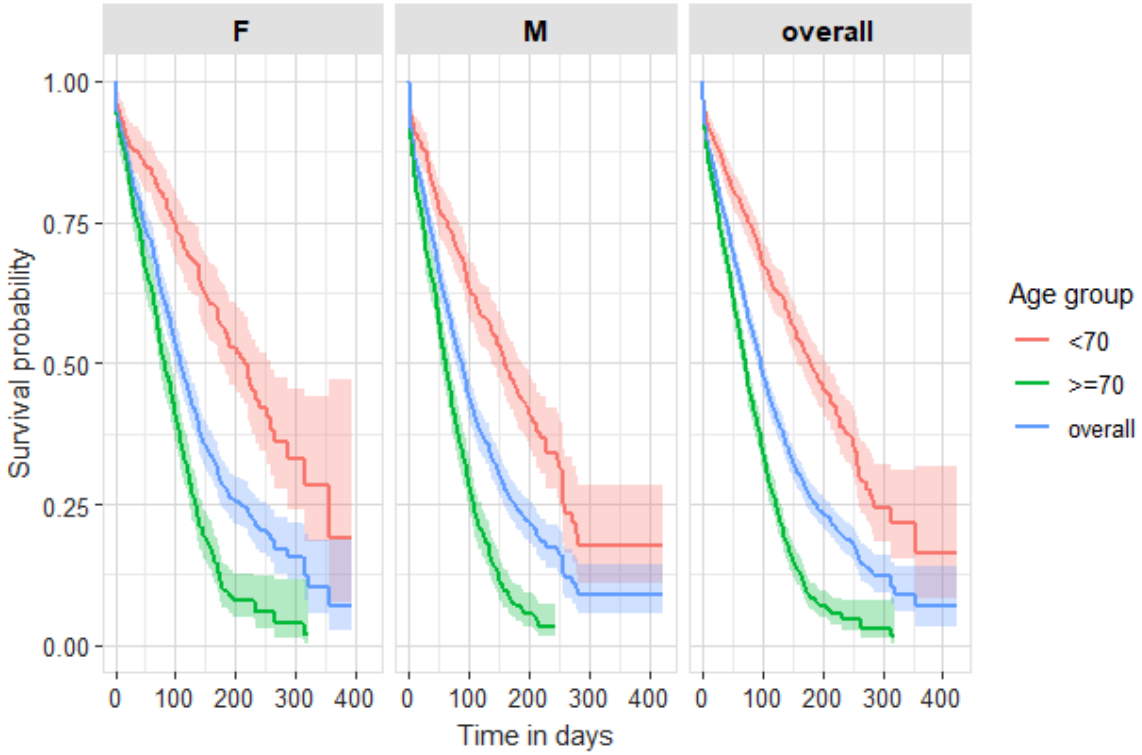
```
> MGUS_death
# A tibble: 10,767 × 13
   result_id cdm_name group_name    group_level    strata_name strata_level variable_name variable_l                    value
       <int> <chr>    <chr>         <chr>          <chr>       <chr>        <chr>         <chr>
 1         1 mock     target_cohort mgus_diagnosis overall     overall      outcome       death_coho
 2         1 mock     target_cohort mgus_diagnosis overall     overall      outcome       death_coho
 3         1 mock     target_cohort mgus_diagnosis overall     overall      outcome       death_coho
 4         1 mock     target_cohort mgus_diagnosis overall     overall      outcome       death_coho
 5         1 mock     target_cohort mgus_diagnosis overall     overall      outcome       death_coho
 6         1 mock     target_cohort mgus_diagnosis overall     overall      outcome       death_coho
 7         1 mock     target_cohort mgus_diagnosis overall     overall      outcome       death_coho
 8         1 mock     target_cohort mgus_diagnosis overall     overall      outcome       death_coho
 9         1 mock     target_cohort mgus_diagnosis overall     overall      outcome       death_coho
10         1 mock     target_cohort mgus_diagnosis overall     overall      outcome       death_cohort
# ℹ 10,757 more rows
# ℹ 2 more variables: additional_name <chr>, additional_level <chr>
```

```
MGUS_death %>%
  distinct(strata_name)
A tibble: 5 × 1
strata_name
<chr>
1 overall
2 age_group
3 sex
4 age_group &&& sex
5 reason
```

curso OMOP

Stratified results

```
plotSurvival(MGUS_death,
             colour = “age_group",
             facet= "sex")
```



curso OMOP

availableSurvivalGrouping

Podemos obtener las variables que se pueden usar en el “facet” y “colour” en los survival plots

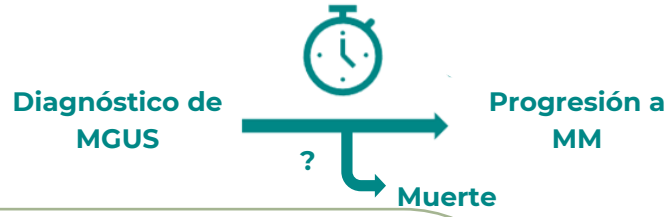
```
availableSurvivalGrouping(MGUS_death)
```

```
[1] "age_group"           "sex"                 "time"                "estimate"
"estimate_95CI_lower" "estimate_95CI_upper"
```

curso OMOP

Competing Risk

Estimar la supervivencia para un evento determinado y el riesgo competitivo




```
estimateCompetingRiskSurvival(
  cdm,
  targetCohortTable,
  targetCohortId = NULL,
  outcomeCohortTable,
  outcomeCohortId = NULL,
  outcomeDateVariable = "cohort_start_date",
  outcomeWashout = Inf,
  competingOutcomeCohortTable,
  competingOutcomeCohortId = NULL,
  competingOutcomeDateVariable = "cohort_start_date",
  competingOutcomeWashout = Inf,
  censorOnCohortExit = FALSE,
  censorOnDate = NULL,
  followUpDays = Inf,
  strata = NULL,
  eventGap = 30,
  estimateGap =1, restrictedMeanFollowUp = NULL, minimumSurvivalDays = 1)
```

curso OMOP

Competing Risk

Ejemplo MGUS

```
MGUS_death_prog <- estimateCompetingRiskSurvival(cdm,
                                                 targetCohortTable = "mgus_diagnosis",
                                                 outcomeCohortTable = "progression",
                                                 competingOutcomeCohortTable = "death_cohort",
                                                 strata = list(c("sex")))
```

```
> MGUS_death_prog
# A tibble: 7,798 × 13
   result_id cdm_name group_name    group_level    strata_name strata_level variable_name variable_level estim
       <int> <chr>    <chr>         <chr>          <chr>       <chr>        <chr>         <chr>          <chr
 1         1 mock     target_cohort mgus_diagnosis overall     overall      outcome       progression    estim
 2         1 mock     target_cohort mgus_diagnosis overall     overall      outcome       progression    estim
 3         1 mock     target_cohort mgus_diagnosis overall     overall      outcome       progression    estim
 4         1 mock     target_cohort mgus_diagnosis overall     overall      outcome       progression    estim
 5         1 mock     target_cohort mgus_diagnosis overall     overall      outcome       progression    estim
 6         1 mock     target_cohort mgus_diagnosis overall     overall      outcome       progression    estim
 7         1 mock     target_cohort mgus_diagnosis overall     overall      outcome       progression    estim
 8         1 mock     target_cohort mgus_diagnosis overall     overall      outcome       progression    estim
 9         1 mock     target_cohort mgus_diagnosis overall     overall      outcome       progression    estim
10         1 mock     target_cohort mgus_diagnosis overall     overall      outcome       progression    estima
# i 7,788 more rows
# i 2 more variables: additional_name <chr>, additional_level <chr>
```

```
> MGUS_death_prog %>%
+     distinct(variable
_level)
# A tibble: 2 × 1
  variable_level
  <chr>
1 progression
2 death_cohort
```

curso OMOP

availableSurvivalGrouping

Podemos obtener las variables que se pueden usar en el “facet” y “colour” en los survival plots

```
availableSurvivalGrouping(MGUS_death_prog)
```

```
[[1] "sex"                  "variable"            "time"                "estimate"
"estimate_95CI_lower"
[6] "estimate_95CI_upper"
```

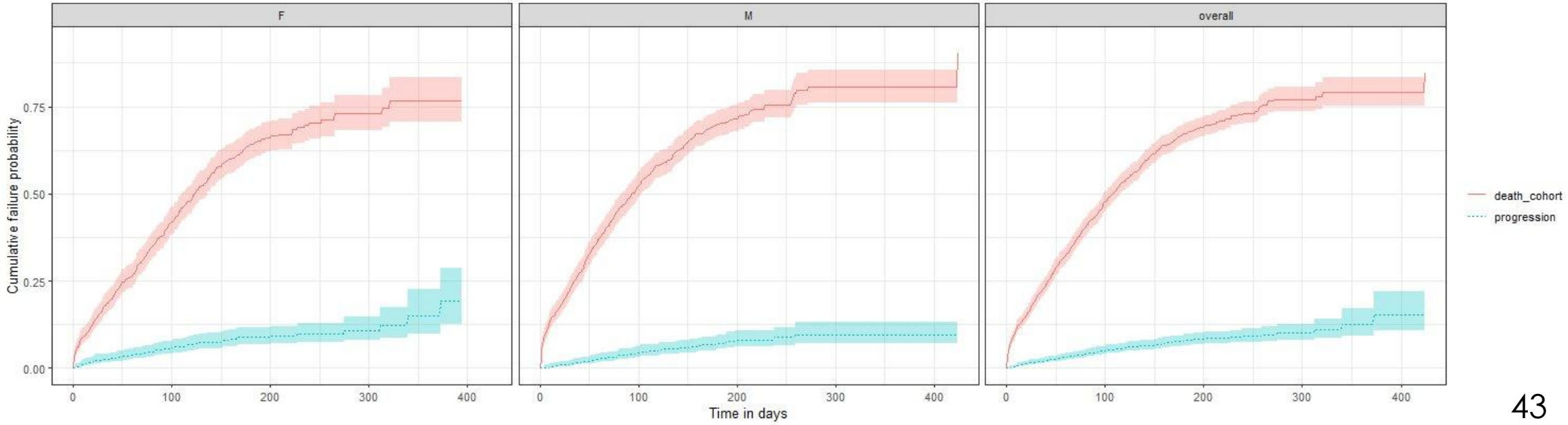
curso OMOP

Competing Risk

Ejemplo MGUS

```
plotSurvival(MGUS_death_prog,
                        cumulativeFailure = TRUE,
                         facet = "sex",
                         colour = "variable")
```

Probabilidad de experimentar un acontecimiento específico en un momento dado, teniendo en cuenta la presencia de acontecimientos de riesgo competitivo



tableSurvival

Resumen de las estimaciones de supervivencia

```
tableSurvival(
X,
times = NULL,
timeScale = “days”,
header = c(“estimate),
estimates = c("median_survival", "restricted_mean_survival"),
type = “gt”,
groupColumn = NULL,
hide = c("result_id", "estimate_type"),
style= NULL,
.options = list()
)
```

curso OMOP

tableSurvival

Resumen de las estimaciones de supervivencia

```
tableSurvival(
x,
times = NULL,
timeScale = “days”,
header = c(“estimate),
estimates = c("median_survival", "restricted_mean_survival"),
type = “gt”,
groupColumn = NULL,
hide = c("result_id", "estimate_type"),
style= NULL,
.options = list()
)
```

Resultados de estimateSingleEventSurvival o estimateCompetingRiskSurvival

curso OMOP

tableSurvival

Resumen de las estimaciones de supervivencia

```
tableSurvival(
x,
times = NULL,
timeScale = “days”,
header = c(“estimate),
estimates = c("median_survival", "restricted_mean_survival"),
type = “gt”,
groupColumn = NULL,
hide = c("result_id", "estimate_type"),
style= NULL,
.options = list()
)
```

Tiempos a los que reportar la supervivencia en la tabla resumen y la escala de dichos tiempos (días, semanas o años)

curso OMOP

tableSurvival

Resumen de las estimaciones de supervivencia

```
tableSurvival(
x,
times = NULL,
timeScale = “days”,
header = c(“estimate),
estimates = c("median_survival", "restricted_mean_survival")
type = “gt”,
groupColumn = NULL,
hide = c("result_id", "estimate_type"),
style= NULL,
.options = list()
)
```

Vector que contiene los elementos que deberían ir en el encabezado. Está permitido que sean cualquiera de los elementos que devuelve la función tableColumns (result)

curso OMOP

tableSurvival

Resumen de las estimaciones de supervivencia

```
tableSurvival(
x,
times = NULL,
timeScale = “days”,
header = c(“estimate),
estimates = c("median_survival", "restricted_mean_survival"),
type = “gt”,
groupColumn = NULL,
hide = c("result_id", "estimate_type"),
style= NULL,
.options = list()
)
```

Vector que contiene las estimaciones que se quieren incluir en la tabla. Las opciones pueden ser: "median_survival", "restricted_mean_survival", "q0_survival", "q05_survival", "q25_survival", "q75_survival", "q95_survival", "q100_survival".

curso OMOP

tableSurvival

Resumen de las estimaciones de supervivencia

```
tableSurvival(MGUS_death)
```

					Estimate name			
CDM name	Target cohort	Age group	Sex	Outcome name	Number records	Number events	Median survival (95% CI)	Restricted mean survival (SE)
mock	mgus_diagnosis	overall	overall	death_cohort	1,384	963	98.00 (92.00, 103.00)	133.00 (4.00)
		<70	overall	death_cohort	574	293	180.00 (158.00, 206.00)	197.00 (8.00)
		>=70	overall	death_cohort	810	670	71.00 (66.00, 77.00)	86.00 (3.00)
		overall	F	death_cohort	631	423	108.00 (100.00, 121.00)	143.00 (6.00)
			M	death_cohort	753	540	88.00 (79.00, 97.00)	125.00 (6.00)
		<70	F	death_cohort	240	109	215.00 (179.00, 260.00)	220.00 (13.00)
			M	death_cohort	334	184	158.00 (139.00, 189.00)	183.00 (10.00)
		>=70	F	death_cohort	391	314	82.00 (75.00, 94.00)	96.00 (4.00)
			M	death_cohort	419	356	61.00 (54.00, 70.00)	80.00 (5.00)

curso OMOP

tableSurvival

Resumen de las estimaciones de supervivencia

```
tableSurvival(MGUS_death, times = c(1, 7, 30, 365), timeScale = “days”)
```

					Estimate name							
CDM name	Target cohort	Age group	Sex	Outcome name	Number records	Number events	Median survival (95% CI)	Restricted mean survival (SE)	1 days survival estimate	7 days survival estimate	30 days survival estimate	365 days survival estimate
mock	mgus_diagnosis	overall	overall	death_cohort	1,384	963	98.00 (92.00, 103.00)	133.00 (4.00)	96.97 (96.07, 97.87)	90.89 (89.39, 92.42)	79.39 (77.28, 81.55)	6.84 (3.36, 13.92)
		<70	overall	death_cohort	574	293	180.00 (158.00, 206.00)	197.00 (8.00)	96.86 (95.45, 98.30)	93.55 (91.57, 95.58)	87.45 (84.78, 90.20)	16.38 (8.43, 31.81)
		>=70	overall	death_cohort	810	670	71.00 (66.00, 77.00)	86.00 (3.00)	97.04 (95.88, 98.21)	89.00 (86.87, 91.18)	73.67 (70.70, 76.77)	-
		overall	F	death_cohort	631	423	108.00 (100.00, 121.00)	143.00 (6.00)	96.83 (95.47, 98.21)	92.70 (90.70, 94.76)	81.89 (78.94, 84.96)	6.90 (2.57, 18.58)
			M	death_cohort	753	540	88.00 (79.00, 97.00)	125.00 (6.00)	97.08 (95.88, 98.29)	89.38 (87.20, 91.60)	77.29 (74.35, 80.34)	8.94 (5.53, 14.44)

curso OMOP

addCohortSurvival

Añade información sobre supervivencia a una cohorte ya existente

```
addCohortSurvival(
  x,
  cdm,
  outcomeCohortTable,
  outcomeCohortId = 1,
  outcomeDateVariable = "cohort_start_date",
  outcomeWashout = Inf,
  censorOnCohortExit = FALSE,
  censorOnDate = NULL,
  followUpDays = Inf,
  name = NULL)
```

Tabla con la cohorte a la que queremos añadir información sobre supervivencia

Se añaden 2 nuevas columnas a la tabla x. Una columna llamada **"days_to_exit"** hasta la salida de la cohorte (por evento o censura). La columna **"status"** indica si el paciente ha tenido el **evento (1) o no (0).**

curso OMOP

addCohortSurvival

Añade información sobre supervivencia a una cohorte ya existente

```
addCohortSurvival(
  x,
  cdm,
  outcomeCohortTable,
  outcomeCohortId = 1,
  outcomeDateVariable = "cohort_start_date",
  outcomeWashout = Inf,
  censorOnCohortExit = FALSE,
  censorOnDate = NULL,
  followUpDays = Inf
  name = NULL)
```

Cohorte con nuestro evento de interés

Se añaden 2 nuevas columnas a la tabla x. Una columna llamada **"days_to_exit"** hasta la salida de la cohorte (por evento o censura). La columna **"status"** indica si el paciente ha tenido el **evento (1) o no (0).**

curso OMOP

addCohortSurvival

Añade información sobre supervivencia a una cohorte ya existente

```
cdm$mgus_diagnosis <- cdm$mgus_diagnosis %>%
  addCohortSurvival(
    cdm = cdm,
    outcomeCohortTable = "death_cohort",
    outcomeCohortId = 1
  )
cdm$mgus_diagnosis
Source:   SQL [?? x 13]
Database: DuckDB v1.1.3 [ilopez@Windows 10 x64:R 4.4.1/:memory:]
```

	cohort_definition_id	subject_id	cohort_start_date	cohort_end_date	age	sex	hgb	creat	mspike	age_group	days_to_exit	status
	<int>	<int>	<date>	<date>	<dbl>	<fct>	<dbl>	<dbl>	<dbl>	<chr>	<int>	<dbl>
1	1	1	1981-01-01	1981-01-01	88	F	13.1	1.3	0.5	>=70	30	1
2	1	2	1968-01-01	1968-01-01	78	F	11.5	1.2	2	>=70	25	1
3	1	3	1980-01-01	1980-01-01	94	M	10.5	1.5	2.6	>=70	46	1
4	1	4	1977-01-01	1977-01-01	68	M	15.2	1.2	1.2	<70	92	1
5	1	5	1973-01-01	1973-01-01	90	F	10.7	0.8	1	>=70	8	1
6	1	6	1990-01-01	1990-01-01	90	M	12.9	1	0.5	>=70	4	1
7	1	7	1974-01-01	1974-01-01	89	F	10.5	0.9	1.3	>=70	151	1
8	1	8	1974-01-01	1974-01-01	87	F	12.3	1.2	1.6	>=70	2	1
9	1	10	1981-01-01	1981-01-01	79	F	9.4	1.1	2.3	>=70	136	1
0	1	11	1972-01-01	1972-01-01	86	M	11.8	1	2.3	>=70	2	1

```
ℹ more rows
ℹ 1 more variable: time <dbl>
```

Resumen

CohortSurvival Package

Resumen de funciones que ofrece el paquete:

- Estimación de supervivencia (un solo acontecimiento): estimateSingleEventSurvival()
- Resumen de los grupos de variables para los gráficos de supervivencia: availableSurvivalGrouping()
- Graficar la supervivencia: plotSurvival()
- Estimación del riesgo competitivo: estimateCompetingRiskSurvival()
- Resumen de estimaciones: tableSurvival()
- Añadir información sobre supervivencia a una cohorte: addCohortSurvival()

Con el output también se pueden usar otros paquetes (ej. para calcular modelos de Cox o log-rank test)

Gracias

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